

Grey Wolf Optimizer (WGO)

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inspiration

Inspired in grey wolf!

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(Click to go the original paper) Published in 2014, in *Advances in engineering*.

Two important aspects

- Social hierarchy
- Hunting mechanism

The ranks as follow:

- *Alpha* (α) wolf
- *Beta* (β) wolf
- *Delta* (δ) wolf
- *Omega* (ω) wolf

Hunting mechanism

The algorithm mainly is based upon the hunting mechanism.

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- Tracking and approaching to the prey.
- Encircling the prey until stop moving.
- Attack the prey.

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The optimization process is guided by the three main wolves (α , β , and δ).

Hunting mechanism

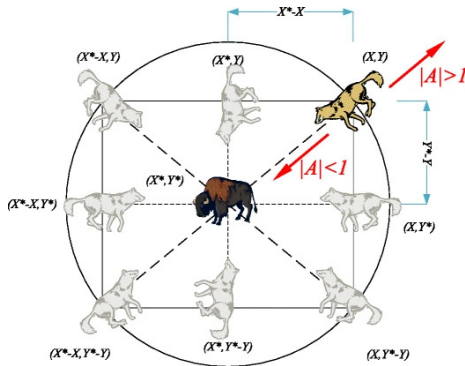


Figure: Extracted from Mirjalili, S. *How effective is the Grey Wolf optimizer in training multi-layer perceptrons.* (2015)

Real life

In real life the position of the pray is unknown therefore the **pack** is guided by the main wolves, the position of dominant wolves is "shared" to the pack (ω wolves).

It is very important *abs()* given that only *A* control the direction of the movement.

Model equations

importante $||$ en WGO not is euclidean distance! is pair wise absolute value.

General idea for a real function $\mathbb{R}^d \rightarrow \mathbb{R}$

Given a problem of d dimension and n agents search (pack size), at t -iteration the matrix of solution is:

$$\underset{n \times d}{\mathbf{X}}^{(t)} \quad (1)$$

fitness

$$f(\mathbf{X}) = f\left(\begin{bmatrix} \vec{x}_1^t \\ \vec{x}_2^t \\ \vdots \\ \vec{x}_n \end{bmatrix}\right) = \begin{bmatrix} f(\vec{x}_1^t) \\ f(\vec{x}_2^t) \\ \vdots \\ f(\vec{x}_n) \end{bmatrix} \quad (2)$$

General idea for a real function $\mathbb{R}^d \rightarrow \mathbb{R}$

Order statistics over fitness values

Take in mind if there is minimization or maximization.

$$f_1 < f_2 < \dots < f_n \quad (3)$$

in the case of minimization then:

$$\alpha = (f_1, \vec{x}_{f_1}^t)$$

$$\beta = (f_2, \vec{x}_{f_2}^t)$$

$$\delta = (f_3, \vec{x}_{f_3}^t)$$

while:

$$\omega_j = (f_j, \vec{x}_{f_j}^t) \quad j = (4, 5, \dots, n)$$

General idea for a real function $\mathbb{R}^n \rightarrow \mathbb{R}$

the update of the $i - th$ solution (wolf) is defined as $\vec{x}_i^{(t+1)}$ where:

$$\vec{x}_i^{(t+1)} = \frac{\vec{\alpha}_i^t + \vec{\beta}_i^t + \vec{\delta}_i^t}{3} \quad \forall i, \text{ where } i = (1, 2, \dots, n) \quad (4)$$

Where the vectors $\vec{\alpha}_i^t + \vec{\beta}_i^t + \vec{\delta}_i^t$ are position vectors of the $i - th$ wolf to each leader respectively.

$$\begin{aligned} \vec{\alpha}_i^t &= \vec{x}_\alpha^t - \vec{A}_\alpha^t \otimes |\vec{c}_\alpha^k \otimes \vec{x}_\alpha^t - \vec{x}_i^t| \\ \vec{\beta}_i^t &= \vec{x}_\beta^t - \vec{A}_\beta^t \otimes |\vec{c}_\beta^k \otimes \vec{x}_\beta^t - \vec{x}_i^t| \\ \vec{\delta}_i^t &= \vec{x}_\delta^t - \vec{A}_\delta^t \otimes |\vec{c}_\delta^k \otimes \vec{x}_\delta^t - \vec{x}_i^t| \end{aligned}$$

remember that for instance $\vec{x}_{f_1}^t = \vec{x}_\alpha^t$.

$$\vec{A}_{rank}^t = 2\vec{a}^t \otimes \vec{r}_1 - \vec{a}^t, \quad rank = (\alpha, \beta, \delta). \quad (5)$$

$$\vec{a}^t = [a^t, \dots, a^t], \quad a^t = 2 - \frac{t}{\text{max_iter}} \quad (6)$$

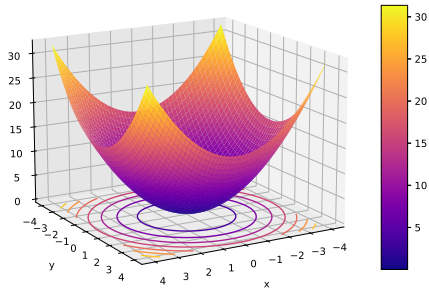
$$\vec{c}_{rank}^t = 2\vec{r}_2, \quad rank = (\alpha, \beta, \delta). \quad (7)$$

$$\vec{r}_i = \text{Uniform}(0, 1), \quad i = (1, 2). \quad (8)$$

vector $\vec{r}$ and $\vec{c}$ are the dimension $(1 \times d)$.

Numerical example

$$f(x, y) = x^2 + y^2$$



Numerical example

$$\min x^2 + y^2 \quad (9)$$

in this problem $d = 2$ y $n = 5$ the pack size, initialize randomly.

$\mathbf{X}^t$	$F(\mathbf{X}^{(t)})$
(2.21, 2.13)	9.43
(1.84, 0.08)	3.4
(0.86, -0.83)	1.42
(1.16, 1.38)	3.23
(0.94, 0.85)	1.61

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$\mathbf{X}^t$	$F(\mathbf{X}^{(t)})$	rank
(0.86, -0.83)	1.42	α
(0.94, 0.85)	1.61	β
(1.16, 1.38)	3.23	δ
(1.84, 0.08)	3.4	ω
(2.21, 2.13)	9.43	ω

Update the first wolf!

 $\vec{A}$

$\vec{x}_1^{t=1} = (2.21, 2.13)$ in each leader wolf $\vec{r}_1$ and $\vec{r}_2$ are defined.

 α

$$\vec{A}_\alpha^{t=1} = (-0.52, 1.8)$$

 β

$$\vec{A}_\beta^{t=1} = (-1.36, -1.36)$$

 δ

$$\vec{A}_\delta^{t=1} = (0.4, 0.84)$$

Update the first wolf!

 $\vec{c}$

$\vec{x}_1^{t=1} = (2.21, 2.13)$ in each leader wolf $\vec{r}_1$ and $\vec{r}_2$ are defined.

 α

$$\vec{c}_\alpha^{t=1} = (1.46, 1.2)$$

 β

$$\vec{c}_\beta^{t=1} = (0.12, 1.74)$$

 δ

$$\vec{c}_\delta^{t=1} = (0.04, 0.94)$$

Update the first wolf

$$\vec{\alpha}_1^{t=1} = \begin{pmatrix} 0.86 \\ -0.83 \end{pmatrix}^T - \begin{pmatrix} -0.52 \\ 1.8 \end{pmatrix}^T \otimes \left| \begin{pmatrix} 1.46 \\ 1.2 \end{pmatrix}^T \otimes \begin{pmatrix} 0.86 \\ -0.83 \end{pmatrix}^T - \begin{pmatrix} 2.21 \\ 2.13 \end{pmatrix}^T \right|$$

$$\vec{\beta}_1^{t=1} = \begin{pmatrix} 0.94 \\ 0.85 \end{pmatrix}^T - \begin{pmatrix} -1.36 \\ -1.36 \end{pmatrix}^T \otimes \left| \begin{pmatrix} 0.12 \\ 1.74 \end{pmatrix}^T \otimes \begin{pmatrix} 0.94 \\ 0.85 \end{pmatrix}^T - \begin{pmatrix} 2.21 \\ 2.13 \end{pmatrix}^T \right|$$

$$\vec{\delta}_1^{t=1} = \begin{pmatrix} 1.16 \\ 1.38 \end{pmatrix}^T - \begin{pmatrix} 0.4 \\ 0.84 \end{pmatrix}^T \otimes \left| \begin{pmatrix} 0.04 \\ 0.94 \end{pmatrix}^T \otimes \begin{pmatrix} 1.16 \\ 1.38 \end{pmatrix}^T - \begin{pmatrix} 2.21 \\ 2.13 \end{pmatrix}^T \right|$$

Therefore

$$\vec{\alpha}_1^{t=1} = \begin{pmatrix} 1.36 \\ -6.46 \end{pmatrix}^T \quad \vec{\beta}_1^{t=1} = \begin{pmatrix} 3.79 \\ 1.74 \end{pmatrix}^T \quad \vec{\delta}_1^{t=1} = \begin{pmatrix} 0.29 \\ 0.68 \end{pmatrix}^T$$

Now we can update the position of the **first wolf** in $\mathbf{X}^{(t+1)}$

$$\begin{aligned} \vec{x}_1^{(t+1)} &= \frac{\vec{\alpha}_1^t + \vec{\beta}_1^t + \vec{\delta}_1^t}{3} \\ &= \begin{pmatrix} 1.81 \\ -1.35 \end{pmatrix}^T \end{aligned}$$

we can make a greedy decision, update only if $f(\vec{x}_1^{(t+1)}) > f(\vec{x}_1^t)$.

Now update $k - th$ wolf

Repeat the update for all wolfs in $\mathbf{X}^{(t)}$ and update t the number of iterations and repeat the process until reach a stopping criteria.

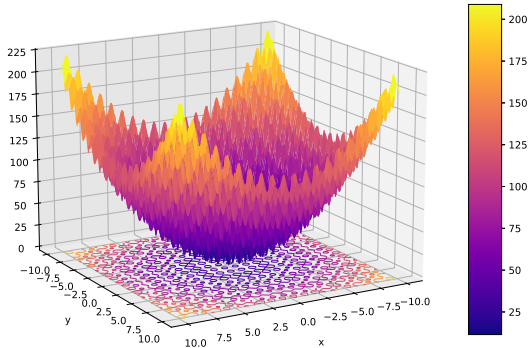
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Lab(numerical example)

Rastrigin function

Have a global minimum but have a lot of local minimuns our algorithm can solve it?

Rastrigin Function in 3D



Solution

$$\operatorname{argmin}_{\vec{x} \in \mathbb{R}^n} f(\vec{x}) = An + \sum_{i=1}^n [x_i^2 - A \cos(2\pi x_i)] \quad (10)$$

$$\vec{x}^* = (0, \dots, 0) \quad (11)$$

...

Lab (rastrigin wolf)